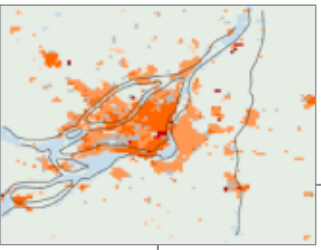
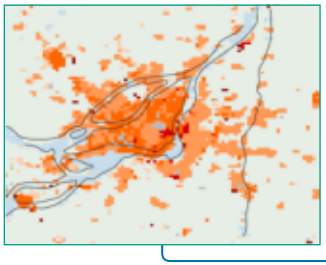
a Static · current practice

Present-day LCZ, held to 2100

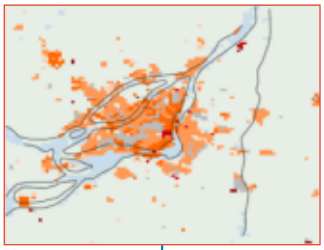


Dynamic · scenario-consistent urban form

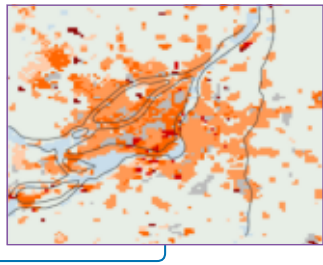
SSP1 · compact, 2100



SSP3 · contraction, 2100

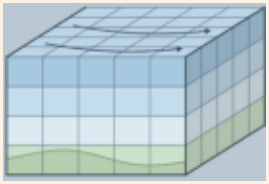


SSP5 · sprawl, 2100



b

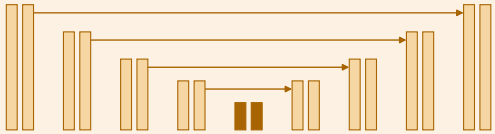
Physics-based downscaling NWP model (e.g. WRF)



Dynamical inference costs
~10^3 × CASPER per period

CASPER · U-Net emulator of WRF

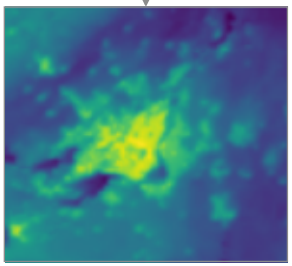
Paired inference: same city, decade, RCP, terrain and atmospheric predictors; only the LCZ-derived land-use input changes.



1-km climate fields
T2 · skin temperature · humidity · wind
6-hourly

3 cities × 11 surface–forcing pairs × 8 decades = 264 runs
~20 min per simulated decade on one GPU

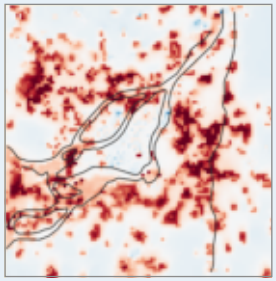
c



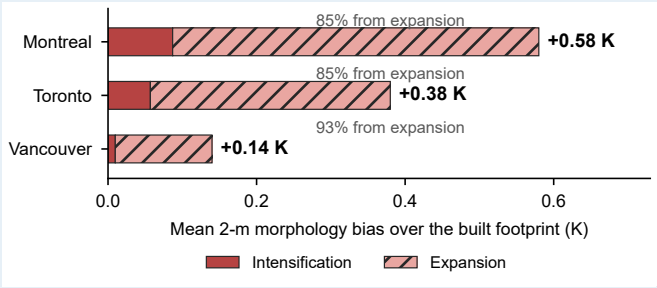
Static-surface temperature Omits the response to evolving urban form

Morphology signal = dynamic − static under matched forcing

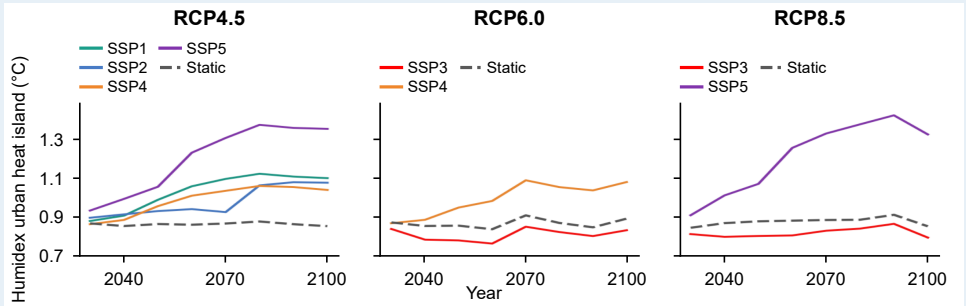
Spatial bias
Montreal · qualitative pattern



Expansion dominates morphological warming
SSP5 · 2100 · area-weighted components



Humidex heat island · Montreal · static baseline dashed



Matched experimental design

8 SSP–RCP combinations
+ static under 3 RCPs

2030–2100 · 3 Canadian cities
Maps shown: Montreal
~1-km grid · 99 × 99 cells

Built LCZ (WUDAPT) 1 2 3 4 5 6 7 8 9 10 · Natural · Water · Coastline

Grey: static · blue: dynamic
Map borders identify SSPs